Smart Suction Cup

Open-Source Tactile Robotic Grasping for Recycling Automation

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BACKGROUND

From a research-grade prototype to an **open-source plug-and-play kit**, we have developed a compact, low-cost, and modular **multi-chambered suction cup end effector** that enables **haptic feedback and automated control** that can be directly deployed on research robots. With future recycling applications in mind, we are making this self-regulating robotic gripper accessible to labs worldwide.

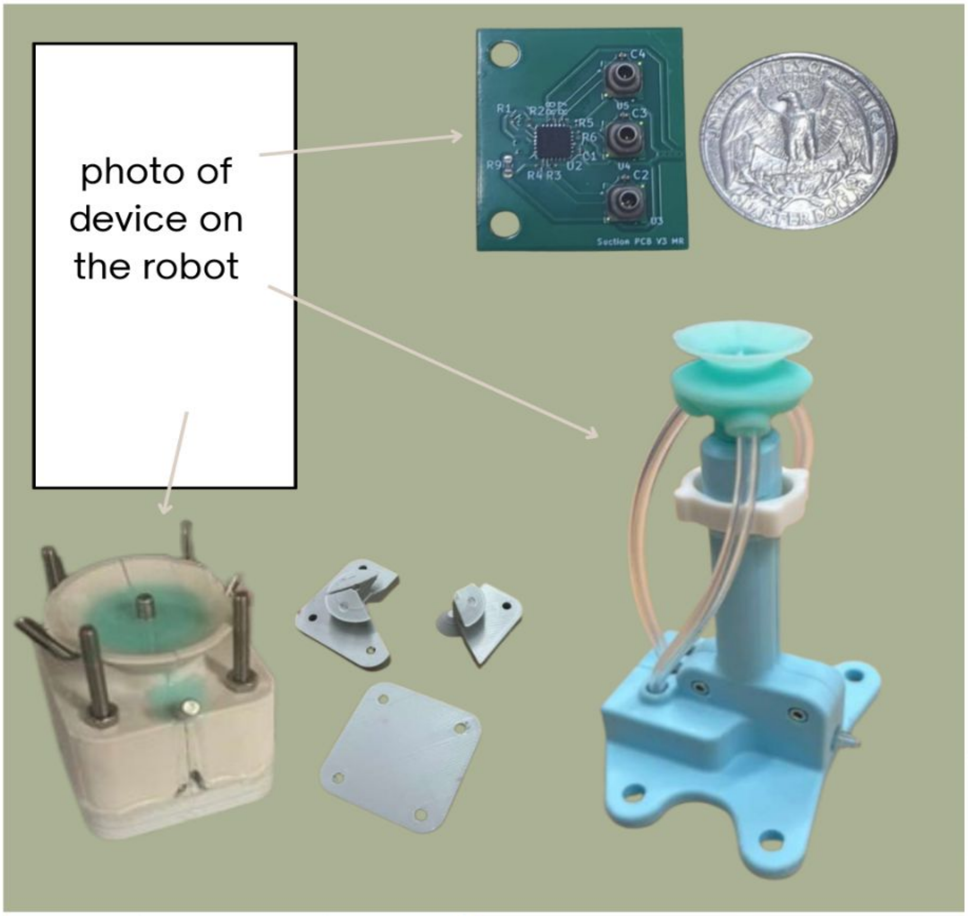
PERFORMANCE TARGETS

- Pick durability: **5000+ cycles**
- Operation time: **60+ minutes**
- Feedback loop: 144 Hz / <10 ms latency
- Suction power: >80% vacuum @ 0.5 kg payload

Price: <$150

EASE OF USE TARGETS

- Compact form factor: **<80×80×60 mm, <1 kg**
- Tool-less cup swap
- Quick setup: **Setup <10 min**
- Open-source & reproducibility



MECHANICAL

CUP & GEOMETRY

- 4-part mold → higher success rate
- Improved geometry for usability

MOUNTING & SWAP

- Intuitive clip-lock mechanism
- ISO 9409-1 compatible casing
- Easy assembly: 2 screws only

Future Improvements

- Internalize exposed tubing
- Add onboard solenoid valve

ELECTRONICS

- From breadboard to integrated PCB
- Quarter-sized (compact)
- Screw-less mounting
- 3 pressure sensors → haptic feedback via differentials

Future Improvements

- Lower price point
- Simplified ordering guide

Metric	Target	Result
Size	≤80×80×60 mm	X mm
Mass	<1 kg	X kg
Payload	0.5 kg	X kg
Vacuum	≥80 %	X %
Lifetime	≥5000 picks	X picks

FUTURE WORK

ADDITIONAL TESTING

- Benchmark vs prior prototypes
- Blind tests by external users

DEPLOYMENT & FAB

- Distribute to other labs
- Explore easier silicone molding
- Low-cost deployment

EDG

Berkeley
Fung Institute for Engineering Leadership